THALLAPALLI INDRASENA REDDY

AI / ML Engineer

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PROFILE SUMMARY

AI/ML Engineer with hands-on experience building enterprise AI systems, RAG-based assistants, and intelligent backend workflows using LLMs, LangGraph, and vector retrieval architectures. Skilled in developing context-aware AI applications, backend APIs, and data-driven solutions across AI, ML, and analytics domains. Focused on building scalable, production-oriented AI systems that solve real-world business problems.

WORK EXPERIENCE

AI / ML Engineer — Onusworks

June 2025 – Present

Bangalore, India (On-site)

AI / ML Development

- Designed and developed VYN Insights, a RAG-based AI assistant for enterprise use cases, enabling contextual and multi-turn query resolution using LLMs.
- Engineered LangGraph-based AI agent architecture with sliding window memory to improve contextual continuity across multi-turn enterprise conversations and query chaining.
- Built LLM-driven MongoDB query generation pipelines enabling natural language interaction with enterprise databases.
- Integrated cost-efficient LLM inference pipelines using LLaMA3 and DeepHermes models via OpenRouter APIs.
- Developed context-aware memory pipelines to improve response relevance and enable seamless conversation continuity.
- Implemented Prompt Engineering techniques to optimize LLM outputs for accuracy and domain-specific relevance.

AI + Data Innovation

- Designed and evaluated predictive ML models for forecasting On-Time In-Full (OTIF) delivery performance using historical logistics and delivery data.
- Integrated Snowflake Cortex Agents for real-time analytics and AI-driven querying directly within the enterprise data warehouse.
- Enabled AI-driven analytics directly within Snowflake, reducing infrastructure complexity.

Data Analytics

- Performed SG&A data analysis, identifying cost drivers and supporting strategic business decisions.
- Collaborated with the ML team to explore predictive models for expense forecasting (research phase).

Backend & Systems

- Developed backend APIs and image-processing workflows for real-time face recognition and identity validation systems.
- Built REST APIs for scalable AI workflow integration.

Cloud & Infrastructure

- Co-engineered Wikihire, an enterprise AI recruitment SaaS platform, building backend routing systems and driving a full infrastructure migration to Google Cloud Platform (GCP).
- Provisioned GCP resources via Terraform IaC and managed IAM-secured service account pipelines across isolated environments.

Machine Learning Engineer (Intern) — VCodez

Feb 2025 – May 2025

Chennai, India

- Spearheaded end-to-end ML model development using Python, Scikit-learn, and Pandas; performed data preprocessing, feature engineering, and model evaluation.
- Developed an AI-powered outfit suggestion system using ML-based image recognition — when a user selects a product, the model identifies the outfit and displays it overlaid on the user's photo.
- Collaborated with cross-functional teams to deploy ML solutions into real-time applications.
- Built insightful data visualizations using Power BI and Excel for stakeholder reporting.
- Mentored junior interns in machine learning, Python programming, and AI tool implementation.
- Effectively communicated technical outcomes to both technical and non-technical stakeholders.

PROJECTS

MAJOR PROJECTS

VYN Insights – Enterprise AI Assistant

Node.js · React · MongoDB · LangChain · OpenAI · LangGraph · Zod · Chart.js · Ant Design · RAG · OpenRouter

- Architected an Autonomous Analytics Agent capable of translating natural language queries into precise MongoDB Aggregation Pipelines (MQL) to analyze enterprise workforce and compliance data end-to-end.

- Implemented a Zero-Trust Schema RAG pipeline where the agent autonomously discovers collections, verifies schema definitions, and validates query syntax before execution — eliminating hallucinated field references.
- Built an Advanced Vector Retrieval System integrating LangChain, MongoDB Vector Search, and OpenAI Embeddings; applied Hypothetical Document Embeddings (HyDE) and an LLM-based Zod-validated refiner to drastically improve schema field precision.
- Enforced multi-tenant data isolation (orgId filtering), strict projection enforcement to prevent context-window bloat, and PII guardrails blocking direct access to sensitive fields (SSN, Aadhaar) while permitting safe existence checks.
- Developed a dynamic React frontend (Ant Design + Chart.js) that adaptively renders AI responses as text summaries, tabular data, or interactive charts (bar, pie, line, scatter) based on query context; integrated speech-to-text input for hands-free analytics.

Vinny – Internal AI Assistant

Node.js · LangChain · MongoDB Vector Search · OpenAI Embeddings · Gemini · JavaScript · RAG · LLMs

- Engineered "Vinny", a context-aware AI chatbot assisting users across Crew and Employee Management workflows within the VYN platform, significantly reducing L2 support ticket volume.
- Designed a 3-layer caching strategy (embedding cache, intent-based cache, and semantic similarity response cache via MongoDB Vector Search) to significantly reduce LLM API costs and response latency.
- Implemented screen-aware contextual AI, making the assistant sensitive to the user's exact application location to provide precise, page-level guidance and improving response relevance.
- Built a deterministic intent classification and escalation system that distinguishes social chit-chat from technical queries and seamlessly routes unsupported cases to human L2 support via strict fallback protocols.
- Engineered a scalable conversation store with automatic TTL background cleanup and per-user thread limits, preventing memory leaks and ensuring efficient resource utilization in production.
- Integrated a user feedback loop (vinny_feedback) to monitor satisfaction signals and continuously improve assistant accuracy over time.

Wikihire – Enterprise AI Recruitment & Requisition SaaS Platform

Node.js · GCP · Terraform · IAM · REST APIs · PostgreSQL · Docker

- Co-engineered an enterprise AI-powered talent acquisition platform featuring a multi-tier role hierarchy (Manager, Department Head, HR) to orchestrate and automate internal hiring approvals and resource requisition pipelines.
- Implemented Infrastructure as Code (IaC) via Terraform to natively provision and manage Google Cloud Platform (GCP) resources, accelerating environment setup and eliminating configuration drift across deployments.
- Managed secure cloud authentication pipelines by deploying GCP Service Account keys across isolated environments under strict IAM principles, ensuring zero-trust access control.
- Optimized core backend API routing architectures to ensure low-latency state persistence during high-volume candidate screening and multi-turn resume intelligence parsing workflows.
- Actively driving a full infrastructure migration of Wikihire to GCP, collaborating with lead engineer to redesign and replatform existing services for cloud-native scalability and resilience.

OTIF Delivery Performance Predictor

Python · ML · Snowflake · Predictive Modeling

- Designed and evaluated predictive ML models for forecasting On-Time In-Full (OTIF) delivery performance using historical logistics and delivery data.
- Integrated Snowflake Cortex Agents for real-time analytics and AI-driven querying directly within the enterprise data warehouse, eliminating additional infrastructure overhead.

Legacy SQL Migration Engine (to BigQuery)

Python · LLMs · BigQuery · SQL Parsing

- Built an LLM-powered SQL migration engine for translating legacy SQL queries into BigQuery-compatible syntax with automated schema and function adaptation.
- Significantly reduced manual effort in DB migration by automating query conversion, enabling teams to migrate large legacy codebases faster and with fewer errors.

ADDITIONAL PROJECTS

Personal RAG Chatbot

Python · LangChain · RAG · FAISS · LLMs

- Built a personal RAG-based chatbot supporting document ingestion, semantic retrieval, and context-aware question answering using FAISS vector search.
- Implemented document chunking, embedding pipelines, and semantic search with FAISS, demonstrating end-to-end understanding of RAG architecture.

Face Recognition System – Backend

Python · Node.js · REST API

- Developed backend APIs and image-processing workflows for real-time face recognition and identity validation systems.
- Integrated image processing pipelines with database matching for high-accuracy identification.

Cyber Attack Breach Prediction

Python · ML · Predictive Modeling

- Developed predictive ML models for cybersecurity breach forecasting and proactive threat risk analysis.
- Improved cybersecurity readiness through proactive threat forecasting and risk mitigation strategies.

SG&A Expense Intelligence & Forecasting

Python · ML · Power BI · Data Analytics

- Performed deep analysis on SG&A expense data to identify key cost drivers, spending patterns, and inefficiencies across business units.
- Collaborated with the ML team to develop predictive models for expense forecasting, enabling data-driven financial planning.

TECHNICAL SKILLS

AI & Machine Learning	Machine Learning, Deep Learning, NLP, LLMs, RAG, AI Agents, Prompt Engineering, Semantic Search
Frameworks & Libraries	LangChain, LangGraph, LlamaIndex, Scikit-learn, Pandas, NumPy
Backend & APIs	Python, Node.js, REST APIs, MongoDB, LLM API Integration
MLOps / LLMOps	Model lifecycle, deployment strategies, monitoring, pipeline design
Data & Analytics	Snowflake, BigQuery, Power BI, Excel
Tools & Platforms	OpenRouter, Groq, GCP, Terraform, Docker, Git, Jupyter Notebook, Agentic IDEs

EDUCATION

Bachelor of Technology in Artificial Intelligence — *Aug 2021 – Mar 2025*
Dr. M.G.R. University, Chennai | **86%**

Intermediate – MPC — *Jun 2019 – May 2021*
Viswa Sai Junior College | State Board | **91.2%**

SSC — *Jun 2018 – Apr 2019*
GSEM High School | State Board of AP | **93%**

CERTIFICATIONS & ACHIEVEMENTS

- Model Context Protocol (MCP) Developer Certification – Anthropic / Claude
- Train and Evaluate Regression Models – Microsoft
- Train and Evaluate Deep Learning Models – Microsoft
- Cybersecurity in a Post-Quantum World – Workshop at SRM University, Chennai